# Project Aevum — Phase 16.3 Freeze Record

VERIFIED & FROZEN 🔒

Independent Facet Appraisal & Preference Formation Engine

| MILESTONE STATUS | TOTAL TESTS PASSED | DESKTOP FRONTEND | DATE & AUTHORITY |
|---|---|---|---|
| FROZEN (Gate 0 + Implementation) | 2,529 / 2,529 (100% Green) | 0 Errors (Vite + TS) | 2026-09-28 / Freeze Gate Passed |

## 1. Executive Summary & North Star Realization

Phase 16.3 marks the critical milestone where Aevum transitions from merely recording experiences to forming **her own authentic, multidimensional, and longitudinal preferences**. Under Phase 16's North Star ("*Aevum does not inherit the user's interests; she develops her own through experience over time*"), taste is neither injected by LLM prompts nor copied from user declarations.

> **Core Epistemic Triad Enforced:**
> - `Knowledge ≠ Experience` (Knowing a game exists is not playing it)
> - `Experience ≠ Preference` (100 grind sessions can yield an `AVOIDED` / `DISLIKED` preference)
> - `User Preference ≠ Aevum Preference` (User enthusiasm cannot fabricate Aevum's affinity)

## 2. Finalized Decisions & Core Architectural Contracts

### Decision 1: Table Naming & Semantic Isolation

To prevent collision with Phase 2 static character definition profiles (which already occupied `character_preferences` in `app/db/models.py`), the Phase 16 preference entity was cleanly named `character_facet_preferences`. This completes a clear architectural progression:

```
character_experiences (Immutable Log) → character_facet_appraisals (Episodic Appraisals) → character_facet_preferences (Materialized Taste)
```

### Decision 2: Source of Truth vs. Materialized Projection

- `character_facet_appraisals` is the **Sole Source of Truth** for episodic evaluations, protected by ORM event listeners (`before_update` and `before_delete` block modifications).
- `character_facet_preferences` is a **Materialized Projection**. It has *zero* ORM immutability constraints. G46 verifies that dropping the preference table and running `rebuild_preferences_from_appraisal_stream()` reproduces the exact identical preference state.

### Decision 3: Permutation & Ingestion Order-Independence (G49)

All consolidation queries order by `timeline_tick ASC`. G49 proves that ingesting appraisals in forward, reversed, or scrambled order produces identical `PersonalPreferenceRecord` projections. The companion's life journey across canonical time dictates taste, never accidental database write ordering.

### Decision 4: 7-Layer Fail-Closed Guard Chain

`FacetAppraisalEngine.evaluate_appraisal()` applies a strict 7-layer defense where any violation immediately fails closed:

| Layer | Guard Check | Violation Error | Constitutional Role |
|---|---|---|---|
| **Guard 1** | Tier-3 User Statement Exclusion | `UngroundedAppraisalEvidenceError` | Anti-Sycophancy: Blocks user recommendations & preferences |
| **Guard 2** | Evidence Status Verification | `UngroundedAppraisalEvidenceError` | Only `GROUNDED` enters engine; rejects `PROPOSED` / `REJECTED` |
| **Guard 3** | Personal Experience Eligibility | `UngroundedAppraisalEvidenceError` | Locks 1 & 9: Requires direct/shared non-fictional interaction |
| **Guard 4** | Domain & Facet Registration | `UnregisteredFacetError` | Prevents orphaned appraisals for unapproved sub-facets |
| **Guard 5** | Quad-Identity Anchoring | `AppraisalIdentityMismatchError` | Enforces identity across character, domain, facet, and experience |
| **Guard 6** | Phase 8C Timeline Anchoring | `TemporalAnchorMismatchError` | `evidence.timeline_tick == record.timeline_tick` (no time drift) |

| Layer | Guard Check | Violation Error | Constitutional Role |
|---|---|---|---|
| **Guard 7** | Idempotency Barrier | `DuplicateAppraisalError` | One experience × one facet → one authoritative appraisal |

## 3. Temperament Semantics & Signal Priority Chain

Temperament is a **deterministic interpretive modifier, not a source of appraisal truth**. It influences interpretation; it does not create reality, evidence, or preference:

| Priority | Signal Condition | Engine Behavior | Verification |
|---|---|---|---|
| **Priority 1** | Unambiguous Signal (e.g. `lyrical_resonance`, `exhausting_loop`) | Direct derivation; **Temperament is completely bypassed** | G42: `lyrical_resonance` + `STRONG_FRICTION` yields `STRONGLY_ATTRACTED` |
| **Priority 2** | Ambiguous Signal (e.g. `routine_interaction`) | Temperament acts as deterministic tie-break | Resonant yields `ENJOYABLE`; friction yields `NEUTRAL_INDIFFERENT` |
| **Priority 3** | Unknown / Missing Signal | Fail-closed (`UnknownGroundedSignalError` / No appraisal) | Engine refuses to guess or invent appraisals from temperament alone |

## 4. Phase 16.2 State Machine Zero Contamination

To ensure strict modularity and protect frozen boundaries:

```
16.2 state_machine.py ←←← [ ZERO IMPORTS FROM 16.3 ] ←←← 16.3 preference_formation_service.py
```

- `PreferenceAwareGrowthService` extends `InterestDomainService` purely via composition.
- `PreferenceCompatibilityProjector` performs one-way translation: `MIXED_AMBIVALENT → "MIXED" → PreferencePolarity.UNFORMED` (never `ATTRACTED`). This preserves 16.2 lifecycle derivation stability.

## 5. Permanent Note A: Zero Floats & Constitutional Audits

- **Zero Floats**: All mathematical thresholds use exact integer ratio math: `(count * 100) ≥ (total * threshold)`. AST recursion confirms 0 float literals or calls.
- **Zero Promotion Authority**: 0 calls to `set_fact()`, 0 uses of `PromotionAuthorizationToken`.
- **Zero Autonomous Agency**: Autonomous curiosity and actions remain strictly locked for Phase 17.

> **Non-Blocking Architectural Note:**
> `character_facet_preferences` is a materialized projection and must never be treated as historical evidence. Any future system inquiring "*Why does Aevum like this?*" must query `character_facet_appraisals`, tracing back to verifiable interaction evidence.

## 6. Complete Verification Audit Matrix

| Test Suite | Tests Executed | Result | Execution Duration |
|---|---|---|---|
| Phase 16.3 Golden Suite (G27–G49) | 26 / 26 | **PASS** (100%) | 0.52s |
| Phase 16 Master Suite (16.1 + 16.2 + 16.3) | 79 / 79 | **PASS** (100%) | 1.37s |
| Monolithic Backend Regression Suite | 2,529 / 2,529 | **PASS** (100%) | 140.81s (2m 20s) |
| Desktop Frontend Production Build | `tsc && vite build` | **PASS** (0 errors) | 821ms |
| Static AST Boundary & Float Audit | Module-wide AST traversal | **PASS** (0 violations) | Instantaneous |

## 7. Roadmap Status & Immediate Next Actions

| Phase | Milestone Name | Status | Key Focus |
|---|---|---|---|
| 16.1 | Experiential Record & Provenance | FROZEN 🔒 | Append-only events, execution evidence anchoring |
| 16.2 | Interest Domain State Machine | FROZEN 🔒 | Lifecycle stage derivation, curiosity authority |
| 16.3 | Facet Appraisal & Preference Formation | **FROZEN** 🔒 | Grounded appraisals, taste formation, G49 invariance |
| **16.4** | **Longitudinal Dynamics: Satiation & Dormancy** | **NEXT →** | Engagement fatigue, novelty decay, resurgence triggers |
| 16.5 | Personal Growth Context Injection | PLANNED | `<personal_growth_context>` XML formatting |
| 17.0 | Autonomous Agency & Curiosity | PLANNED | Independent companion agency during schedule ticks |